The finite-element calculation used by the circuit model

1 Problem Setup

Our goal is to solve $\nabla \cdot \boldsymbol{a} = 0$ over domain Ω, with $\boldsymbol{a} = c(\nabla u - \boldsymbol{f})$. For our specific setup, u is the electric potential. For the conduction problem inside the conductors, $\boldsymbol{a} = -\boldsymbol{J}$, c is the conductivity, and $\boldsymbol{f}$ is the driving field within the battery. For the electrostatics problem outside the conductors, $c = 1$, $\boldsymbol{f} = 0$.

The problem domain Ω is first divided into triangles. For each vertex i, define $N_i(\boldsymbol{x})$ as a basis function that takes value 1 at vertex i and value 0 at all other vertices. Between vertices, the function is piecewise linear. Over domain Ω, we assume

$$u(\boldsymbol{x}) = \sum_i u_i N_i(\boldsymbol{x}), \tag{1}$$

where u_i is the value of the potential at vertex i. We may know the potential values at a subset of vertices; the values u_i at the remaining vertices are unknowns we need to solve.

2 Turning field equations into a linear system

To derive a linear system, multiply $\nabla \cdot \boldsymbol{a} = 0$ by a function w and use the product rule and the divergence theorem:

$$\begin{aligned} 0 &= \int_\Omega w \, \nabla \cdot \boldsymbol{a} \, dA && (2)\\ &= \int_\Omega \nabla \cdot (w\boldsymbol{a}) \, dA - \int_\Omega \boldsymbol{a} \cdot \nabla w \, dA && (3)\\ &= \int_{\partial\Omega} w \, \boldsymbol{a} \cdot \boldsymbol{n} \, ds - \int_\Omega \boldsymbol{a} \cdot \nabla w \, dA. && (4) \end{aligned}$$

After substituting $\boldsymbol{a}$, this is

$$\int_\Omega c \, \nabla u \cdot \nabla w \, dA = \int_\Omega c \, \boldsymbol{f} \cdot \nabla w \, dA + \int_{\partial\Omega} w \, \boldsymbol{a} \cdot \boldsymbol{n} \, ds. \tag{5}$$

Let's focus on $\int_{\partial\Omega} w \, \boldsymbol{a} \cdot \boldsymbol{n} \, ds$. If (1) $w = 0$ on the boundary $\partial\Omega$, or (2) $\boldsymbol{a} \cdot \boldsymbol{n} = 0$ on the boundary $\partial\Omega$, this term will vanish. Assume w is chosen such that either (1) or (2) is true, we have

$$\int_\Omega c \, \nabla u \cdot \nabla w \, dA = \int_\Omega c \, \boldsymbol{f} \cdot \nabla w \, dA. \tag{6}$$

The remaining question is how to choose w. We choose $w(x) = N_i(\boldsymbol{x})$ for vertex i that is not on $\partial\Omega$, and for vertex i that is on $\partial\Omega$ if we have $\boldsymbol{a} \cdot \boldsymbol{n} = 0$ on $\partial\Omega$. If vertex i is not on $\partial\Omega$, from the definition of N_i, we know $N_i(\boldsymbol{x}) = 0$ on $\partial\Omega$, so the condition (1) above is satisfied. For a chosen vertex that is on $\partial\Omega$, the condition (2) above is satisfied. Thus Equation 6 holds for our chosen w.

3 The linear system in discrete form

We turn the integral in Equation 6 into a summation over triangles in the mesh:

$$\sum_h \int_{\Omega_h} c_h \, \nabla u_h \cdot \nabla w_h \, dA = \sum_h \int_{\Omega_h} c_h \, \boldsymbol{f}_h \cdot \nabla w_h \, dA. \tag{7}$$

We label triangle h's vertices as "$h, 0$", "$h, 1$", and "$h, 2$", and get

$$u_h(\boldsymbol{x}) = u_{h,0} N_{h,0}(\boldsymbol{x}) + u_{h,1} N_{h,1}(\boldsymbol{x}) + u_{h,2} N_{h,2}(\boldsymbol{x}). \tag{8}$$

Substituting this expression into Equation 7, we get

$$\sum_h \sum_{k=0}^{2} \int_{\Omega_h} c_h \, u_{h,k} \nabla N_{h,k} \cdot \nabla w_h \, dA = \sum_h \int_{\Omega_h} c_h \, \boldsymbol{f}_h \cdot \nabla w_h \, dA. \tag{9}$$

In our problems, the integrands above are constant within each triangle h. Writing its area as A_h, we get

$$\sum_h \sum_{k=0}^{2} c_h A_h \, (\nabla N_{h,k} \cdot \nabla w_h) \, u_{h,k} = \sum_h c_h A_h \, (\boldsymbol{f}_h \cdot \nabla w_h). \tag{10}$$

This is a linear system in discrete form, where the subset of $u_{h,k}$ that are unknown are variables to solve, and the number of equations is the number of different w functions that we choose according the rules specified above.

4 Gradients on a triangle

To keep the notation simplified, we hide the index h, and consider a generic triangle with vertices $\boldsymbol{x}_0$, $\boldsymbol{x}_1$, and $\boldsymbol{x}_2$. The three corresponding basis functions are $N_0(\boldsymbol{x})$, $N_1(\boldsymbol{x})$, $N_2(\boldsymbol{x})$.

Set $\boldsymbol{p} = \boldsymbol{x}_1 - \boldsymbol{x}_0$, $\boldsymbol{q} = \boldsymbol{x}_2 - \boldsymbol{x}_0$, and $D = p_x q_y - p_y q_x$. The triangle area is $A = |D|/2$. The basis gradients are

$$\nabla N_1 = \frac{(q_y, -q_x)}{D}, \qquad \nabla N_2 = \frac{(-p_y, p_x)}{D}, \qquad \nabla N_0 = -\nabla N_1 - \nabla N_2. \tag{11}$$

The signed D gives the same physical gradients whether the vertices are listed clockwise or counterclockwise.

Code: `calculations/fem.py`.